Incidence Geometry with Polarity and Tiled Surfaces

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1 Introduction

Fomin and Pylyavskyy introduced a tiling framework for projective incidence geometry based on quadrangulations [2]. Points and lines label the two vertex classes, and each face carries a cross-ratio whose value measures its coherence. On a closed oriented surface, these local weights satisfy a global product identity: every edge contributes to two adjacent faces with opposite exponents, so all edge contributions cancel. As a consequence, coherence of all but one face forces coherence of the remaining face.

The aim of this paper is to bring projective duality directly into this cancellation mechanism. We fix a nondegenerate symmetric bilinear form β on a three-dimensional vector space over a field $\Bbbk$, and use the resulting polarity to replace each line label by its pole. This turns an FP tiling into a tiling labeled entirely by projective points. The conversion preserves the cross-ratio of every face, and hence preserves both face weights and coherence. It also gives projective duality a concrete form within the point-labeled model: duality exchanges the two pairs of opposite labels on a polarity tile while leaving its weight unchanged.

The point-labeled formulation also suggests how the cancellation argument can survive on a nonorientable surface. In the usual FP setting, the point-line coloring is globally consistent because every closed edge path has even length. On a nonorientable surface, traveling around a closed path may reverse the local orientation, and the two vertex types should then be exchanged. The relevant condition is therefore a compatibility between edge parity and orientation. For a quadrangulation Q of a closed surface S, the parity of closed edge paths determines an $\mathbb{F}_2$-valued cohomology class, while the first Stiefel–Whitney class $w_{1,S}$ records whether those paths preserve or reverse orientation. We call Q *orientation-twisted* when these two classes agree. Equivalently, orientation-preserving closed edge paths have even length, and orientation-reversing ones have odd length.

On the orientation double cover, an orientation-twisted quadrangulation becomes bipartite, and the deck involution exchanges the two vertex colors. This is precisely the same color exchange that represents projective duality after polarity conversion. The local edge cancellations can therefore be carried out on the cover and then descended to the original surface. When S is orientable, the condition reduces to the usual bipartiteness of $Q^{(1)}$. We use the standard properties of the first Stiefel–Whitney class and the orientation double cover recorded in [5, 3].

These observations lead to the polarity version of the FP product identity, which is the main result of this paper.

Theorem 1.1 (Polarity Master Theorem)**.** *Let $Q^{\mathcal{L}}$ be an orientation-twisted polarity tiling of a closed connected surface S, and assume that every edge pairing is nonzero.*

If S is orientable, choose an orientation of S and a two-coloring of $Q^{(1)}$, and read $\chi_\beta(f)$ from the positively oriented boundary of f, beginning at a color-zero vertex. If S is nonorientable, define $\chi_\beta(f)$ using the orientation-cover lift. Then

$$\prod_{f \in F(Q)} \chi_\beta(f) = 1.$$

Consequently, if every face except a marked face f_0 is coherent, then f_0 is coherent.

For a nonorientable surface, the lifted quadrangulation on the orientation double cover is bipartite, and the deck involution exchanges its two colors. The deck involution also reverses the orientation of each lifted face and therefore exchanges the two opposite pairs in the polarity cross-ratio. The symmetry of χ_β makes the face weight independent of the chosen lift. The product identity then follows from the usual local cancellation along every edge.

The orientation cover also relates polarity tilings to self-dual incidence theorems. A strictly orientation-twisted polarity tiling lifts canonically to a bipartite FP tiling on which the deck involution acts as projective duality. Conversely, a bipartite quadrangulation with an admissible color-swapping involution descends to an orientation-twisted polarity tiling. The Hesse–Desargues configuration gives the principal example of this lift [1]. The bordered form of the master theorem produces coherent polarity polygons and the Pascal configuration, following the coherent-polygon construction of Fomin and Pylyavskyy [2, Proposition 4.2 and Corollary 4.3].

We also develop two extensions of the polarity master theorem. Using the gauge-theoretic correspondence of Izosimov [4], we associate a group-valued connection with a σ-Hermitian polarity over a division ring. Face coherence is identified with trivial facial holonomy, leading to a Hermitian polarity master theorem under the corresponding algebraic and topological hypotheses. We then study the least Euler genus of an elementary orientation-twisted tiling proof. The resulting characterization uses parity-refined commutator and square factorizations in the premise group and extends Zhang's orientable genus formula to the nonorientable setting [7].

Sections 2 and 3 establish the projective and topological background, introduce polarity conversion, and show that it preserves coherence and projective duality. Pascal's theorem is presented as an example of the FP tiling framework, while Brianchon's theorem illustrates how projective duality acts after polarity conversion.

Section 4 introduces orientation-twisted quadrangulations, characterizes them through the orientation double cover, and proves the Polarity Master Theorem. It then discusses the bordered case and presents the Hesse and Klein bottle polarity configurations.

Section 5 studies the relation between orientation covers and self-dual incidence theorems. It constructs the canonical bipartite lift of a polarity tiling, explains the Hesse–Desargues example, and characterizes the reverse descent in terms of admissible color-swapping involutions.

Section 6 develops the division-ring extension. It reformulates coherence as flatness of a group-valued connection, constructs the connection associated with a σ-Hermitian polarity, proves the Hermitian Polarity Master Theorem, and gives a complex Hermitian obstruction to the Hesse analogue.

Finally, Section 7 studies the topology of elementary orientation-twisted tiling proofs. It introduces parity-refined commutator and square lengths, treats the orientable and nonorientable cases separately, and derives the elementary Euler-genus formula.

2 Preliminaries

Throughout this paper, let $\Bbbk$ be a field and let V be a three-dimensional vector space over $\Bbbk$. We fix a non-degenerate symmetric bilinear form

$$\beta\colon V\times V\longrightarrow \Bbbk.$$

Definition 2.1 (Projective points, lines, and incidence)**.** We write $\mathbb{P}(V)$ for the projective plane of one-dimensional subspaces of V. For a projective point $A\in\mathbb{P}(V)$, choose a nonzero vector $a\in V$ representing A and write

$$A=[a].$$

A projective line is identified with a point of $\mathbb{P}(V^*)$. If $\alpha\in V^*\setminus\{0\}$, the corresponding projective line is

$$\ell_\alpha := \mathbb{P}(\ker\alpha).$$

For $A=[a]$ and $\ell=\ell_\alpha$, we write

$$\langle A,\ell\rangle := \alpha(a).$$

Although this scalar depends on the chosen representatives a and α, its vanishing does not:

$$\langle A,\ell\rangle = 0 \quad\Longleftrightarrow\quad A\in\ell.$$

2.1 Projective Polarities

Definition 2.2 (Orthogonal complement)**.** For a vector subspace $U\subseteq V$, define

$$U^\perp := \{x\in V : \beta(u,x)=0 \text{ for every } u\in U\}.$$

Definition 2.3 (Projective polarity)**.** Let $A=[a]\in\mathbb{P}(V)$. The *polar line* of A is

$$A^\perp := \mathbb{P}\big((\Bbbk a)^\perp\big) = \{[x]\in\mathbb{P}(V) : \beta(a,x)=0\}.$$

Let $L=\mathbb{P}(W)$ be a projective line, where $\dim W=2$. The *pole* of L is

$$L^\perp := \mathbb{P}(W^\perp).$$

Remark 2.4 (Well-definedness). These definitions are independent of the chosen vector representatives. The non-degeneracy of β gives

$$\dim(\Bbbk a)^\perp = 2,\qquad \dim W^\perp = 1,$$

so $A^\perp$ is a projective line and $L^\perp$ is a projective point.

Proposition 2.5 (Polarity reciprocity)**.** *For every projective point A and every projective line L,*

$$A\in L \quad\Longleftrightarrow\quad L^\perp\in A^\perp.$$

Moreover,

$$(A^\perp)^\perp = A,\qquad (L^\perp)^\perp = L.$$

Proof. Write $A = [a]$ and $L = \mathbb{P}(W)$. Then

$$A \in L \iff \Bbbk a \subseteq W \iff W^\perp \subseteq (\Bbbk a)^\perp.$$

Projectivizing the last inclusion gives

$$L^\perp \in A^\perp.$$

The remaining identities follow from

$$(U^\perp)^\perp = U,$$

which holds for every subspace $U \subseteq V$ because β is non-degenerate. □

Definition 2.6 (β-conjugacy)**.** Two projective points $A = [a]$ and $B = [b]$ are called *β-conjugate* if

$$\beta(a, b) = 0.$$

Equivalently,

$$B \in A^\perp.$$

Two projective lines L and M are called *β-conjugate* if

$$L^\perp \in M.$$

By polarity reciprocity, this is equivalent to

$$M^\perp \in L.$$

We write

$$L \perp_\beta M.$$

Remark 2.7 (Conjugate lines determined by point pairs). For distinct projective points A and B,

$$(AB)^\perp = A^\perp \cap B^\perp.$$

Consequently, for distinct $A, B, C, D \in \mathbb{P}(V)$,

$$AB \perp_\beta CD \iff (AB)^\perp \in CD \iff (CD)^\perp \in AB.$$

2.2 Fomin–Pylyavskyy Tiles

We recall the incidence-tiling framework of Fomin and Pylyavskyy [2]. Their local object is a quadrilateral whose two classes of opposite vertices carry point labels and line labels, respectively.

Definition 2.8 (Fomin–Pylyavskyy tile)**.** A *Fomin–Pylyavskyy tile*, or an *FP tile*, is a topological quadrilateral whose boundary vertices are labeled in cyclic order by

$$A - \ell - B - m - A,$$

where $A, B \in \mathbb{P}(V)$ are projective points and $\ell, m \subseteq \mathbb{P}(V)$ are projective lines. We denote this tile by

$$S(A, B; \ell, m).$$

The tile is called *admissible* if neither A nor B lies on either ℓ or m. Equivalently,

$$\langle A, \ell\rangle\langle B, \ell\rangle\langle B, m\rangle\langle A, m\rangle \neq 0.$$

The tile is called *degenerate* if

$$A = B \qquad \text{or} \qquad \ell = m,$$

and *nondegenerate* otherwise.

An admissible FP tile is called *coherent* if it is degenerate, or if it is nondegenerate and the three lines

$$AB, \qquad \ell, \qquad m$$

are concurrent.

Remark 2.9 (Collinearity form of coherence). For a nondegenerate FP tile, coherence is equivalently expressed by the collinearity condition

$$A, \qquad B, \qquad \ell \cap m \quad \text{are collinear.}$$

(a) Boundary order of an FP tile.

(b) Coherence: AB, ℓ, m are concurrent.

Figure 1: An FP tile and its geometric coherence condition in the projective plane. Black vertices carry point labels, while white vertices carry line labels.

Remark 2.10 (Two descriptions of FP coherence). Figure 1 displays the two descriptions of coherence. The cyclic order

$$A - \ell - B - m - A$$

records the combinatorial tile, while the intersection

$$AB \cap \ell \cap m$$

records its geometric realization.

Definition 2.11 (Mixed cross-ratio)**.** The *mixed cross-ratio* of an admissible FP tile is

$$\chi(A, B; \ell, m) := \frac{\langle A, \ell\rangle\langle B, m\rangle}{\langle A, m\rangle\langle B, \ell\rangle}.$$

Remark 2.12 (Symmetry of the mixed cross-ratio). The mixed cross-ratio is independent of the chosen vector and covector representatives. It also satisfies

$$\chi(A, B; \ell, m) = \chi(B, A; m, \ell).$$

Proposition 2.13 (Cross-ratio criterion for coherence)**.** *An admissible FP tile $S(A,B;\ell,m)$ is coherent if and only if*

$$\chi(A,B;\ell,m)=1.$$

Proof. If $A=B$ or $\ell=m$, the equality follows directly from the definition of the mixed cross-ratio.

Suppose that the tile is nondegenerate. Write

$$A=[a], \qquad B=[b],$$

and represent ℓ and m by nonzero covectors

$$\alpha,\mu\in V^*,$$

respectively.

A point of AB has a representative

$$x=sa+tb, \qquad (s,t)\neq(0,0).$$

The lines AB,ℓ,m are concurrent precisely when there is a nonzero pair (s,t) such that

$$\alpha(x)=0, \qquad \mu(x)=0.$$

Equivalently,

$$\begin{pmatrix}\alpha(a) & \alpha(b)\\ \mu(a) & \mu(b)\end{pmatrix}\begin{pmatrix}s\\ t\end{pmatrix}=\begin{pmatrix}0\\ 0\end{pmatrix}.$$

Such a nonzero solution exists if and only if

$$\alpha(a)\mu(b)-\alpha(b)\mu(a)=0.$$

Since the tile is admissible, all four evaluations are nonzero. Therefore the last equality is equivalent to

$$\frac{\alpha(a)\mu(b)}{\mu(a)\alpha(b)}=1,$$

which is precisely

$$\chi(A,B;\ell,m)=1.$$

□

2.3 Incidence Tilings

Definition 2.14 (Quadrangular complex)**.** A *quadrangular complex* is a finite two-dimensional regular cell complex whose 2-cells are quadrilaterals. Its sets of vertices, edges, and faces are denoted by

$$V(Q), \qquad E(Q), \qquad F(Q).$$

Definition 2.15 (Quadrangulation)**.** Let S be a compact surface, possibly with boundary. A quadrangular complex Q is called a *quadrangulation* of S if its cells form a cell decomposition of S.

Equivalently, every interior edge is incident with exactly two faces, every boundary edge is incident with exactly one face, the link of every interior vertex is a circle, and the link of every boundary vertex is an interval.

Remark 2.16 (Specified face structure). A quadrangulation includes its specified face structure. In particular, a 4-cycle in $Q^{(1)}$ need not be the boundary of a face.

Definition 2.17 (Bipartite labeling and FP tiling)**.** A *bipartite labeling* of a quadrangular complex Q is a map

$$\mathcal{B}\colon V(Q) \longrightarrow \mathbb{F}_2 \times \mathbb{N}$$

such that adjacent vertices have different first components. Thus

$$\mathrm{pr}_1\big(\mathcal{B}(u)\big) \neq \mathrm{pr}_1\big(\mathcal{B}(v)\big)$$

for every edge $uv \in E(Q)$.

A quadrangular complex equipped with a bipartite labeling is called an *FP tiling* and is denoted by

$$Q^{\mathcal{B}}.$$

Definition 2.18 (Projective realization)**.** The first component of $\mathcal{B}$ records the type of the label. We use the convention

$$(0,n) \quad \text{for a point label } P_n, \qquad (1,n) \quad \text{for a line label } \ell_n.$$

Consequently, point and line labels alternate around the boundary of every face.

A *projective realization* of $Q^{\mathcal{B}}$ assigns a projective point P_n to every point label $(0,n)$ and a projective line ℓ_n to every line label $(1,n)$. The realization is called *edge-admissible* if the point and line assigned to the endpoints of every edge are nonincident.

Definition 2.19 (Face weight and coherent face)**.** Let $Q^{\mathcal{B}}$ be an edge-admissible FP tiling of an oriented surface S, and let $f \in F(Q)$. Orient f by the orientation of S, and begin its boundary at a point vertex. If the resulting cyclic boundary is

$$A - \ell - B - m - A,$$

then f determines the FP tile

$$S(A, B; \ell, m)$$

and the face weight

$$\chi(f) := \chi(A, B; \ell, m).$$

The face f is called *coherent* if its associated FP tile is coherent.

Theorem 2.20 (Fomin–Pylyavskyy master theorem)**.** *Let $Q^{\mathcal{B}}$ be an edge-admissible FP tiling of a closed connected oriented surface S. Then*

$$\prod_{f \in F(Q)} \chi(f) = 1.$$

Consequently, if every face except one is coherent, then the remaining face is coherent as well.

Proof. For a positively oriented face boundary

$$A - \ell - B - m - A,$$

the corresponding weight is

$$\chi(f) = \frac{\langle A, \ell\rangle\langle B, m\rangle}{\langle A, m\rangle\langle B, \ell\rangle}.$$

An edge pairing occurs with exponent $+1$ when the oriented boundary traverses the edge from a point vertex to a line vertex, and with exponent -1 when it traverses the edge from a line vertex to a point vertex.

Every edge of a closed surface is incident with exactly two faces. These faces induce opposite orientations on their common edge. Therefore every edge pairing occurs once with exponent $+1$ and once with exponent -1 in

$$\prod_{f \in F(Q)} \chi(f).$$

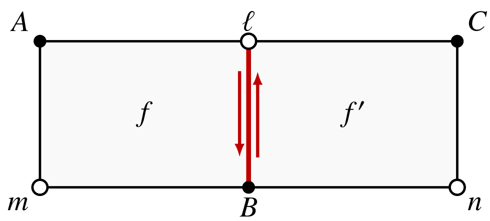
All edge pairings cancel, giving

$$\prod_{f \in F(Q)} \chi(f) = 1.$$

By Proposition 2.13, a face is coherent if and only if its weight is 1. Hence, if every face except f_0 is coherent, then

$$1 = \prod_{f \in F(Q)} \chi(f) = \chi(f_0),$$

so f_0 is coherent. □



$$\chi(f) = \frac{\langle A, \ell\rangle\langle B, m\rangle}{\langle A, m\rangle\langle B, \ell\rangle}, \quad \chi(f') = \frac{\langle B, \ell\rangle\langle C, n\rangle}{\langle B, n\rangle\langle C, \ell\rangle}.$$

Figure 2: Local cancellation along a shared edge. The pairing $\langle B, \ell\rangle$ occurs once in a denominator and once in a numerator because the two faces induce opposite directions on their common edge.

Definition 2.21 (Tiled incidence implication)**.** Let $Q^{\mathcal{B}}_{f_0}$ be an edge-admissible FP tiling with a distinguished face f_0. For every face f, let $I(f)$ denote the assertion that f is coherent. The marked tiling carries the incidence implication

$$\bigwedge_{f \in F(Q)\setminus\{f_0\}} I(f) \quad \Longrightarrow \quad I(f_0).$$

Remark 2.22 (Incidence theorems carried by marked tilings)*.* When Q is a closed connected oriented surface, the validity of this implication follows from Theorem 2.20. In this sense, the tiling gives a topological proof of the incidence theorem carried by its marked face.

Fomin and Pylyavskyy show that many classical results of projective incidence geometry, including the theorems of Desargues, Pappus, Möbius, and the complete quadrangle theorem, arise from such marked tilings [2].

2.4 Topological Conventions

Remark 2.23 (Surface and cell-structure conventions). All surfaces considered in this paper are compact and connected unless stated otherwise. A cell structure is always assumed to be finite and regular. Thus every closed cell is embedded, and the intersection of two closed cells is either empty or a common face of both.

Definition 2.24 (Orientation character)**.** Let S be a surface. We write

$$w_{1,S} \in H^1(S; \mathbb{F}_2)$$

for its first Stiefel–Whitney class. For every closed path γ in S,

$$\langle w_{1,S}, [\gamma] \rangle = \begin{cases} 0, & \gamma \text{ preserves local orientation,} \\ 1, & \gamma \text{ reverses local orientation.} \end{cases}$$

In particular,

$$w_{1,S} = 0 \quad \Longleftrightarrow \quad S \text{ is orientable.}$$

Definition 2.25 (Orientation double cover)**.** The orientation double cover of S is denoted by

$$\pi \colon \widetilde{S} \longrightarrow S,$$

and its deck involution by

$$\tau \colon \widetilde{S} \longrightarrow \widetilde{S}.$$

When S is nonorientable, $\widetilde{S}$ is connected and τ is fixed-point-free and orientation-reversing. Moreover,

$$\pi \circ \tau = \pi.$$

If Q is a cell structure on S, its orientation-cover lift is

$$\widetilde{Q} := \pi^{-1}(Q).$$

Every cell of Q has two lifts, and the deck involution exchanges them.

2.5 Premise Complexes

Definition 2.26 (Fixed-form incidence implication)**.** We now record the topological construction associated with a fixed-form incidence implication. Let $\mathcal{P}$ be a finite set of point variables and let $\mathcal{H}$ be a finite set of line variables. For $i = 0, 1, \ldots, m$, let

$$Q_i = S(A_i, B_i; \ell_i, m_i),$$

where

$$A_i, B_i \in \mathcal{P}, \qquad \ell_i, m_i \in \mathcal{H}.$$

The boundary of Q_i is the cyclic word

$$\partial Q_i = A_i - \ell_i - B_i - m_i - A_i.$$

We write

$$I\colon Q_1, \dots, Q_m \Longrightarrow Q_0,$$

where $Q_1, \dots, Q_m$ are the premise squares and Q_0 is the conclusion square. We assume that Q_0 is nondegenerate.

At this stage, the arrow records the formal arrangement of the premises and conclusion. A projective realization turns each Q_i into the coherence statement associated with the corresponding FP tile.

Definition 2.27 (Incidence graph and premise complex)**.** The *incidence graph* Γ_I has one vertex for every point or line variable appearing in $Q_0, Q_1, \dots, Q_m$. Two vertices are joined by an edge whenever the corresponding variables occur consecutively on the boundary of one of these squares.

Choose an orientation on every edge of Γ_I. Reading the boundary of Q_i gives a closed edge word

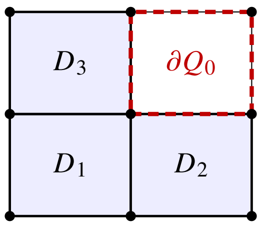
$$\partial Q_i,$$

where an edge is written with exponent -1 whenever it is traversed against the chosen orientation.

For each premise square Q_i, attach a square 2-cell D_i along ∂Q_i. The resulting square complex

$$X_I := \Gamma_I \cup_{\partial Q_1} D_1 \cup \cdots \cup_{\partial Q_m} D_m$$

is called the *premise complex* of I.



$$X_I = \Gamma_I \cup_{\partial Q_1} D_1 \cup_{\partial Q_2} D_2 \cup_{\partial Q_3} D_3.$$

Figure 3: A schematic premise complex. The premise squares are attached as 2-cells, while the boundary of the conclusion square remains a distinguished loop representing the conclusion class δ_I.

Definition 2.28 (Premise group and conclusion class)**.** The conclusion boundary ∂Q_0 is a closed edge-path in $\Gamma_I \subseteq X_I$. Let X_I^0 be the connected component of X_I containing this path, and define the *premise group* by

$$G_I := \pi_1(X_I^0).$$

After choosing a basepoint on ∂Q_0, the conclusion boundary represents an element of G_I. Changing the basepoint conjugates this element, while reversing the boundary direction replaces it by its inverse. We denote the resulting unoriented conjugacy class by

$$\delta_I$$

and call it the *conclusion class* of I.

Remark 2.29 (The unattached conclusion boundary). Figure 3 emphasizes that the conclusion square is not attached to X_I. Only its boundary is retained, and this boundary determines the conclusion class δ_I.

2.6 The Parity Character

Definition 2.30 (Parity character)**.** Assign the value $1 \in \mathbb{F}_2$ to every edge of X_I^0. This defines a cellular 1-cochain

$$\lambda_I \in C^1(X_I^0; \mathbb{F}_2).$$

Every 2-cell of X_I^0 comes from a premise square and has four boundary edges. Hence

$$d\lambda_I = 0,$$

so λ_I determines a cohomology class

$$[\lambda_I] \in H^1(X_I^0; \mathbb{F}_2).$$

Equivalently, this class determines a homomorphism

$$\varepsilon_I \colon G_I \longrightarrow \mathbb{F}_2,$$

characterized by

$$\varepsilon_I([\gamma]) = |\gamma| \pmod 2$$

for every closed edge-path γ in X_I^0. We call ε_I the *parity character* of I.

Since the conclusion boundary has four edges,

$$\varepsilon_I(\delta_I) = 0.$$

Remark 2.31 (Pullback interpretation of the parity character). Let S be a compact surface equipped with a square cell structure, and let

$$f \colon S \longrightarrow X_I^0$$

be a cellular map that sends every square of S homeomorphically onto one of the premise cells $D_1, \dots, D_m$. Then

$$f^*[\lambda_I] \in H^1(S; \mathbb{F}_2)$$

records the parity of closed edge-paths in the induced cell structure.

The relation

$$f^*[\lambda_I] = w_{1,S}$$

is equivalent to

$$\varepsilon_I \circ f_* = w_{1,S}.$$

This formulation will later allow the parity condition to be expressed through the orientation-cover lift.

3 From Incidence Tilings to Polarity Tilings

We now pass from the point-line labeling used by Fomin and Pylyavskyy to a labeling entirely by projective points. The fixed polarity identifies every projective line with its pole, thereby converting an FP tile into a polarity tile. This conversion preserves the cross-ratio, coherence, and projective duality.

3.1 The Polarity Cross-Ratio and Polarity Tiles

Definition 3.1 (Polarity cross-ratio)**.** Let

$$A = [a], \qquad B = [b], \qquad C = [c], \qquad D = [d] \in \mathbb{P}(V),$$

and suppose that

$$\beta(a, c), \qquad \beta(b, c), \qquad \beta(b, d), \qquad \beta(a, d)$$

are all nonzero. The *polarity cross-ratio* of the ordered quadruple $(A, B; C, D)$ is

$$\chi_\beta(A, B; C, D) := \frac{\beta(a, c)\beta(b, d)}{\beta(a, d)\beta(b, c)}.$$

Remark 3.2 (Symmetries of the polarity cross-ratio). The polarity cross-ratio is independent of the chosen vector representatives. The symmetry of β also gives

$$\chi_\beta(A, B; C, D) = \chi_\beta(B, A; D, C) = \chi_\beta(C, D; A, B).$$

Definition 3.3 (Polarity tile)**.** A *polarity tile* is a closed oriented disk with four marked boundary vertices labeled in positive cyclic order by

$$A - C - B - D - A.$$

It is denoted by

$$(A, B; C, D),$$

so A, B form one pair of opposite labels and C, D form the other.

The tile is called *edge-admissible* if adjacent labels are not β-conjugate; equivalently,

$$\beta(a, c)\beta(b, c)\beta(b, d)\beta(a, d) \neq 0.$$

The tile is called *degenerate* if

$$A = B \qquad \text{or} \qquad C = D,$$

and *nondegenerate* otherwise.

An edge-admissible polarity tile is called *coherent* if

$$\chi_\beta(A,B;C,D) = 1.$$

Remark 3.4 (Degenerate polarity tiles). In a degenerate polarity tile, the four marked boundary vertices remain distinct; only two opposite vertices carry the same label.

Lemma 3.5 (Coherence of degenerate tiles)**.** *Every degenerate edge-admissible polarity tile is coherent.*

Proof. If $A = B$, choose the same representative a for the two occurrences of A. Then

$$\chi_\beta(A,A;C,D) = \frac{\beta(a,c)\beta(a,d)}{\beta(a,d)\beta(a,c)} = 1.$$

The case $C = D$ is analogous. ◻

Proposition 3.6 (Geometric characterization)**.** *An edge-admissible polarity tile* $(A,B;C,D)$ *is coherent if and only if it is degenerate, or it is nondegenerate and*

$$AB \perp_\beta CD.$$

In the nondegenerate case,

$$\chi_\beta(A,B;C,D) = 1 \quad \Longleftrightarrow \quad (AB)^\perp \in CD \quad \Longleftrightarrow \quad (CD)^\perp \in AB.$$

Equivalently,

$$(AB)^\perp,\ C,\ D \quad \textit{are collinear},$$

and

$$A,\ B,\ (CD)^\perp \quad \textit{are collinear}.$$

The same condition is equivalent to the concurrence of

$$AB,\ C^\perp,\ D^\perp,$$

and to the concurrence of

$$A^\perp,\ B^\perp,\ CD.$$

Proof. The degenerate case follows from Lemma 3.5. Suppose that

$$A \neq B \qquad \text{and} \qquad C \neq D.$$

A point of CD has a representative

$$x = sc + td, \qquad (s,t) \neq (0,0).$$

It lies in

$$(AB)^\perp = A^\perp \cap B^\perp$$

precisely when

$$\beta(a,x) = 0, \qquad \beta(b,x) = 0.$$

Equivalently,
$$\begin{pmatrix} \beta(a,c) & \beta(a,d) \\ \beta(b,c) & \beta(b,d) \end{pmatrix} \begin{pmatrix} s \\ t \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$
A nonzero solution exists if and only if
$$\beta(a,c)\beta(b,d) - \beta(a,d)\beta(b,c) = 0.$$
Since the four boundary pairings are nonzero, this is equivalent to
$$\chi_\beta(A,B;C,D) = 1.$$
Therefore
$$\chi_\beta(A,B;C,D) = 1 \quad \Longleftrightarrow \quad (AB)^\perp \in CD.$$
Polarity reciprocity gives
$$(AB)^\perp \in CD \quad \Longleftrightarrow \quad (CD)^\perp \in AB.$$
Finally,
$$(CD)^\perp = C^\perp \cap D^\perp, \qquad (AB)^\perp = A^\perp \cap B^\perp,$$
which gives the collinearity and concurrency formulations. □

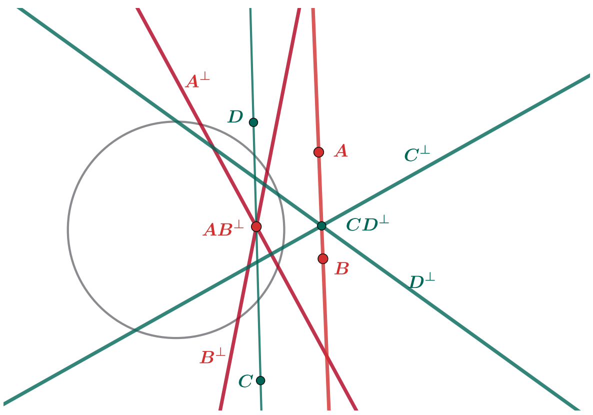


Figure 4: Geometric realization of a coherent polarity tile. The picture illustrates the equivalent conditions in Proposition 3.6, including the collinearity of $A, B, (CD)^\perp$.

3.2 Polarity Tilings

Let Q be a quadrangular complex as in Definition 2.14.

Definition 3.7 (Polarity labeling)**.** A *polarity labeling* of Q is a map
$$\mathcal{L} \colon V(Q) \longrightarrow \mathbb{P}(V).$$
A quadrangular complex equipped with a polarity labeling is called a *polarity tiling* and is denoted by
$$Q^{\mathcal{L}}.$$

The polarity tiling is called *edge-admissible* if

$$\beta(a_u, a_v) \neq 0$$

for every edge $uv \in E(Q)$, where

$$\mathcal{L}(u) = [a_u], \qquad \mathcal{L}(v) = [a_v].$$

Definition 3.8 (Face tile and coherent face)**.** Let $f \in F(Q)$, and choose an orientation of f. Write its boundary in positive cyclic order as

$$v_1 - v_2 - v_3 - v_4 - v_1.$$

Then f determines the polarity tile

$$\big(\mathcal{L}(v_1), \mathcal{L}(v_3); \mathcal{L}(v_2), \mathcal{L}(v_4)\big).$$

Reversing the orientation of f replaces its polarity cross-ratio by its inverse. Therefore the condition that f is coherent is independent of the chosen orientation.

Definition 3.9 (Marked polarity tiling)**.** If f is distinguished, we write

$$Q_f^{\mathcal{L}} := (Q, f, \mathcal{L})$$

and call it a *marked polarity tiling*. For two distinguished faces f and g, we write

$$Q_{f,g}^{\mathcal{L}} := (Q, f, g, \mathcal{L}).$$

3.3 From FP Tilings to Polarity Tilings

Definition 3.10 (Polarity conversion of a labeling)**.** Let $Q^{\mathcal{B}}$ be a projectively realized FP tiling. Fix the polarity induced by β. Define

$$\psi_\beta$$

on the point and line labels of the realization by

$$\psi_\beta(P_n) := P_n, \qquad \psi_\beta(\ell_n) := \ell_n^\perp.$$

Thus point labels remain unchanged, while line labels are replaced by their poles.

The resulting polarity labeling

$$\mathcal{L}_\beta \colon V(Q) \longrightarrow \mathbb{P}(V)$$

is given by

$$\mathcal{L}_\beta(v) := \begin{cases} P_n, & \mathcal{B}(v) = (0, n), \\ \ell_n^\perp, & \mathcal{B}(v) = (1, n). \end{cases}$$

We denote the associated polarity tiling by

$$Q^{\mathcal{L}_\beta}.$$

Consider a face of $Q^{\mathcal{B}}$ with cyclic boundary

$$A - \ell - B - m - A.$$

Set

$$C := \ell^{\perp}, \qquad D := m^{\perp}.$$

The corresponding face of $Q^{\mathcal{L}_\beta}$ has cyclic boundary

$$A - C - B - D - A$$

and determines the polarity tile

$$(A, B; C, D).$$

Choose nonzero vectors $c, d \in V$ representing C, D. Since

$$\ell = C^{\perp}, \qquad m = D^{\perp},$$

the covectors representing ℓ and m are proportional to

$$x \longmapsto \beta(x, c), \qquad x \longmapsto \beta(x, d),$$

respectively. It follows that

$$\chi(A, B; \ell, m) = \frac{\beta(a, c)\beta(b, d)}{\beta(a, d)\beta(b, c)} = \chi_\beta(A, B; C, D).$$

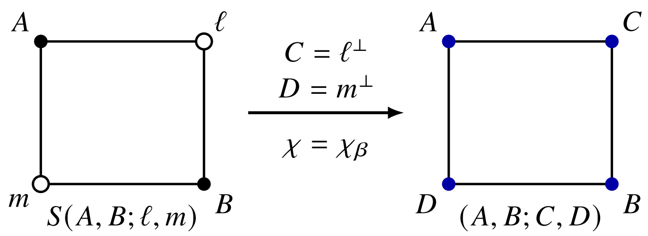


Figure 5: Polarity conversion of a face. Each line label is replaced by its pole, and the mixed cross-ratio becomes the polarity cross-ratio.

Remark 3.11 (Cyclic order under polarity conversion). Figure 5 shows that polarity conversion changes only the type of the line labels. The underlying quadrilateral and its cyclic order are preserved:

$$A - \ell - B - m - A \quad \longmapsto \quad A - \ell^{\perp} - B - m^{\perp} - A.$$

Proposition 3.12 (Polarity conversion)**.** *Replacing every line label of an edge-admissible FP tiling by its pole produces an edge-admissible polarity tiling. This conversion preserves the cross-ratio and coherence of every face.*

Conversely, suppose that $Q^{(1)}$ is bipartite and let

$$c\colon V(Q) \longrightarrow \mathbb{F}_2$$

be a two-coloring. A polarity labeling

$$\mathcal{L}\colon V(Q) \longrightarrow \mathbb{P}(V)$$

determines an FP realization by assigning

$$P_v := \mathcal{L}(v) \qquad \textit{when } c(v) = 0,$$

and

$$\ell_v := \mathcal{L}(v)^{\perp} \qquad \textit{when } c(v) = 1.$$

These two constructions are inverse to one another.

Proof. The equality

$$\chi(A, B; \ell, m) = \chi_\beta(A, B; \ell^\perp, m^\perp)$$

shows that the face weights and coherence conditions are preserved. The two constructions are inverse because polarity reciprocity gives

$$(P^\perp)^\perp = P, \qquad (\ell^\perp)^\perp = \ell.$$

□

Corollary 3.13 (Marked polarity conversion)**.** *In particular, a marked FP tiling*

$$Q_f^{\mathcal{B}}$$

gives a marked polarity tiling

$$Q_f^{\mathcal{L}_\beta}$$

with the same premise and conclusion face relations.

Example 3.14 (Pascal's theorem)**.** Let

$$P_1, P_2, P_3, P_4, P_5, P_6$$

be six points on a nondegenerate conic. Set

$$P_7 = (P_1P_2) \cap (P_4P_5), \quad P_8 = (P_2P_3) \cap (P_5P_6),$$
$$P_9 = (P_3P_4) \cap (P_6P_1).$$

Pascal's theorem states that

$$P_7,\ P_8,\ P_9 \quad \text{are collinear.}$$

We use this classical theorem as an example of the coherent-polygon construction of Fomin and Pylyavskyy [2, Proposition 4.2 and Corollary 4.3].

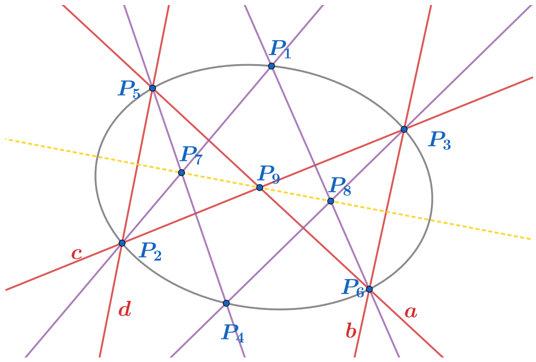


Figure 6: The Pascal configuration. The points P_7, P_8, P_9 lie on the dashed Pascal line.

Tiling proof. Consider the coherent octagon

$$\mathcal{O} = P_1 - c - P_4 - b - P_1 - a - P_4 - d - P_1$$

appearing on the left of Figure 7. By the coherent-polygon construction, $\mathcal{O}$ admits a filling by coherent quadrilateral tiles.

The disk on the right has the same boundary and contains the additional point labels P_7 and P_8. By their defining incidence relations, all its quadrilateral faces are coherent except possibly the central face

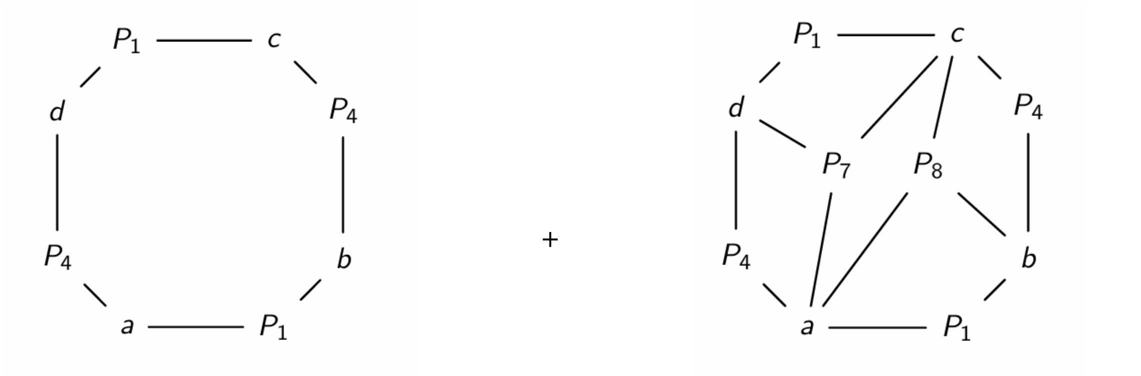
$$S(P_7, P_8; c, a).$$

Glue a coherent filling of the left-hand octagon to the right-hand disk along their common boundary. The result is an FP tiling of the sphere in which only the central face remains to be checked. By Theorem 2.20, this face is coherent.

Proposition 2.13 now shows that P_7P_8, c, a are concurrent. Since the conic construction gives $c \cap a = P_9$, we obtain

$$P_9 \in P_7P_8.$$

Thus P_7, P_8, P_9 are collinear. $\square$



(a) The coherent boundary octagon.

(b) The disk containing P_7, P_8.

Figure 7: The tiling proof of Pascal's theorem. After gluing along the common boundary, the master theorem forces the central face to be coherent.

3.4 Polarity Conversion and Projective Duality

Definition 3.15 (Color-swapping duality)**.** Projective duality exchanges point and line labels:

$$P_n \longmapsto P_n^{\perp}, \qquad \ell_n \longmapsto \ell_n^{\perp}.$$

After polarity conversion, the projective labels remain unchanged, while the two vertex types are exchanged. Indeed, a point label transforms as

$$P_n \longmapsto P_n^{\perp} \longmapsto (P_n^{\perp})^{\perp} = P_n,$$

while a line label transforms as

$$\ell_n \longmapsto \ell_n^{\perp}.$$

Define

$$\varphi\colon \mathbb{F}_2 \times \mathbb{N} \longrightarrow \mathbb{F}_2 \times \mathbb{N}, \qquad \varphi(b,n) = (b+1,n),$$

and write

$$\varphi\mathcal{B} := \varphi \circ \mathcal{B}.$$

Under polarity conversion, projective duality is represented by the color-swapping operation

$$Q^{\mathcal{B}} \longmapsto Q^{\varphi\mathcal{B}}.$$

Proposition 3.16 (Duality compatibility)**.** *For a face with converted polarity labels $(A,B;C,D)$, projective duality exchanges the two opposite pairs. The identity*

$$\chi_{\beta}(A,B;C,D) = \chi_{\beta}(C,D;A,B)$$

shows that the face weight is invariant under this operation. Consequently, polarity conversion is compatible with projective duality.

Example 3.17 (Brianchon's theorem)**.** Let

$$\ell_1, \ell_2, \ldots, \ell_6$$

be six tangent lines to a nondegenerate conic, written in cyclic order. Define

$$\begin{aligned}
\ell_7 &= \big(\ell_1 \cap \ell_2\big)\big(\ell_4 \cap \ell_5\big),\\
\ell_8 &= \big(\ell_2 \cap \ell_3\big)\big(\ell_5 \cap \ell_6\big),\\
\ell_9 &= \big(\ell_3 \cap \ell_4\big)\big(\ell_6 \cap \ell_1\big),
\end{aligned}$$

where XY denotes the line through two points X and Y. Brianchon's theorem states that

$$\ell_7,\ \ell_8,\ \ell_9 \quad \text{are concurrent.}$$

It is the projective dual of Pascal's theorem.

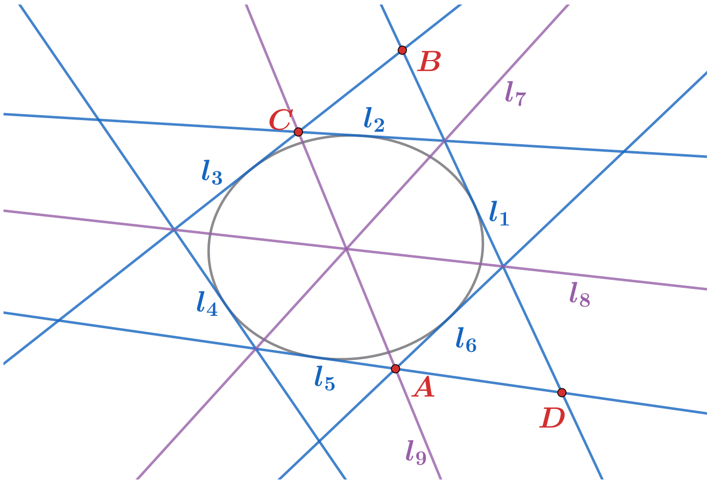


Figure 8: The Brianchon configuration. The three lines ℓ_7, ℓ_8, ℓ_9 meet at a common point.

Proof by projective duality. Apply the polarity induced by the conic to the Pascal configuration in Example 3.14. Each point P_i on the conic is sent to its tangent line

$$\ell_i = P_i^{\perp}.$$

For example,

$$\begin{aligned} P_7^{\perp} &= \big((P_1P_2) \cap (P_4P_5)\big)^{\perp} \\ &= \big(\ell_1 \cap \ell_2\big)\big(\ell_4 \cap \ell_5\big) = \ell_7. \end{aligned}$$

Similarly,

$$P_8^{\perp} = \ell_8, \qquad P_9^{\perp} = \ell_9.$$

Pascal's theorem gives a line p containing P_7, P_8, P_9. Polarity reciprocity therefore gives

$$p^{\perp} \in P_7^{\perp} \cap P_8^{\perp} \cap P_9^{\perp} = \ell_7 \cap \ell_8 \cap \ell_9.$$

Hence ℓ_7, ℓ_8, ℓ_9 are concurrent.

At the level of the tiling, projective duality replaces

$$P_1, P_4, P_7, P_8 \quad \text{by} \quad \ell_1, \ell_4, \ell_7, \ell_8,$$

and replaces

$$a, b, c, d \quad \text{by} \quad A, B, C, D.$$

The underlying quadrangular complex and its filling pattern are unchanged, as shown in Figure 9. □

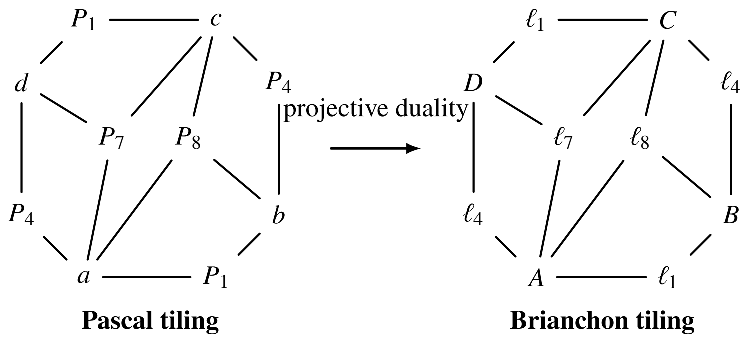


Figure 9: Projective duality exchanges the point and line labels of the Pascal tiling while preserving its underlying quadrangular filling. The result is the Brianchon tiling.

Remark 3.18 (Duality and the orientation-cover coloring). The next section introduces the orientation-twisted condition. For a strictly orientation-twisted polarity tiling, the orientation-cover lift admits a two-coloring satisfying

$$c \circ \tau = c + 1.$$

Thus the deck involution acts as the same color-swapping operation that represents projective duality in the FP model.

4 Polarity Master Theorem

4.1 Orientation-Twisted Quadrangulation

Definition 4.1 (Cohomological characterization)**.** Define the all-ones cochain

$$\lambda \in C^1(Q; \mathbb{F}_2), \qquad \lambda(e) = 1,$$

for every edge e. Because for every face f, the cellular coboundary is $(d\lambda)(f) = \lambda(\partial f) = 0 \in \mathbb{F}_2$, the cochain λ is a cocycle. Its class $\ell = [\lambda] \in H^1(S; \mathbb{F}_2)$ satisfies

$$\langle \ell, [\gamma] \rangle = |\gamma| \pmod 2$$

for every closed edge path γ.

Moreover, a function $c \in C^0(Q; \mathbb{F}_2)$ is a two-coloring of the one-skeleton if and only if $dc = \lambda$.

Definition 4.2 (Orientation-twisted quadrangulation)**.** The quadrangulation Q of S is *orientation-twisted* if $\ell = w_{1,S}$. It is *strictly orientation-twisted* if, in addition, S is nonorientable.

Intuitively, a quadrangulation is orientation-twisted if and only if every orientation-preserving closed edge path has even length and every orientation-reversing closed edge path has odd length. In particular, when S is orientable, orientation-twisted means precisely that $Q^{(1)}$ is bipartite. We use the standard characteristic-class and orientation-cover facts recorded in [5, 3].

Recall that the first Stiefel-Whitney class w_1 is the obstruction to orientability, which evaluates to zero on orientation-preserving loops and to one on orientation-reversing loops. We can construct it in Q as following. Let $S = |Q|$, and let $p : \mathrm{Or}(S) \to S$ be the orientation double cover with deck involution τ. Choose one lift $\widetilde{v} \in p^{-1}(v)$ for every vertex v. For an oriented edge $e : u \to v$, lift e

from $\widetilde{u}$ and define $\varepsilon(e) \in \mathbb{F}_2$ by requiring its endpoint to be $\tau^{\varepsilon(e)}\widetilde{v}$. Since the boundary of every face is null-homotopic, $\mathrm{d}\varepsilon = 0$. Replacing $\widetilde{v}$ by $\tau^{s(v)}\widetilde{v}$ changes ε to $\varepsilon + \mathrm{d}s$; hence $[\varepsilon]$ is independent of all choices. Its evaluation on a closed edge path records whether parallel transport preserves or reverses local orientation. Therefore $[\varepsilon] = w_1(S)$.

Proposition 4.3 (Orientation-cover coloring criterion)**.** *Let Q be a quadrangulation of a closed connected nonorientable surface S. Let $\pi : \widetilde{S} \to S$ be the orientation double cover, let $\widetilde{Q} = \pi^{-1}(Q)$, and let τ be the deck involution. Then Q is orientation-twisted if and only if $\widetilde{Q}$ is bipartite and admits a two-coloring $c : V(\widetilde{Q}) \to \mathbb{F}_2$ satisfying $c \circ \tau = c + 1$.*

Proof. Let $\lambda \in Z^1(Q; \mathbb{F}_2)$ be the edge-length cocycle, $\lambda(e) = 1$, and write $\ell = [\lambda]$.

Suppose first that Q is orientation-twisted, so that $\ell = w_{1,S}$. Naturality of the first Stiefel–Whitney class gives

$$\pi^*\ell = \pi^* w_{1,S} = w_{1,\widetilde{S}} = 0.$$

Hence $\pi^*\lambda = dc$ for some $c \in C^0(\widetilde{Q}; \mathbb{F}_2)$. Since $\pi^*\lambda(\widetilde{e}) = 1$ for every edge $\widetilde{e}$ of $\widetilde{Q}$, the function c changes value across every edge and therefore defines a two-coloring of $\widetilde{Q}$.

Choose a path $\widetilde{\gamma}$ from a vertex $\widetilde{v}$ to $\tau\widetilde{v}$. Its projection $\gamma = \pi \circ \widetilde{\gamma}$ is an orientation-reversing closed edge path in S. Therefore

$$|\widetilde{\gamma}| = |\gamma| = \langle \ell, [\gamma] \rangle = \langle w_{1,S}, [\gamma] \rangle = 1 \pmod 2.$$

Since c changes along every edge, it follows that $c(\tau\widetilde{v}) = c(\widetilde{v}) + 1$. Moreover,

$$d(c \circ \tau + c) = \tau^*(dc) + dc = \tau^*(\pi^*\lambda) + \pi^*\lambda = 0.$$

Thus $c \circ \tau + c$ is constant on the connected graph $\widetilde{Q}^{(1)}$. Since its value at $\widetilde{v}$ is 1, we obtain $c \circ \tau = c + 1$ at every vertex.

Conversely, suppose that $\widetilde{Q}$ admits a two-coloring c satisfying $c \circ \tau = c + 1$. Let γ be any closed edge path in Q, and lift it to a path $\widetilde{\gamma}$ beginning at $\widetilde{v}$. If γ is orientation-preserving, then $\widetilde{\gamma}$ ends at $\widetilde{v}$, so its length is even. If γ is orientation-reversing, then $\widetilde{\gamma}$ ends at $\tau\widetilde{v}$; since these endpoints have opposite colors, its length is odd. Consequently,

$$\langle \ell, [\gamma] \rangle = |\gamma| = \langle w_{1,S}, [\gamma] \rangle \pmod 2$$

for every closed edge path γ. Hence $\ell = w_{1,S}$, and Q is orientation-twisted. □

4.2 Polarity Master Theorem

Lemma 4.4 (Quotient polarity cross ratio)**.** *Let $Q^{\mathcal{L}}$ be a strictly orientation-twisted polarity tiling, and let $(\widetilde{Q}, \tau)$ be its orientation-cover lift. Assume that every edge pairing is nonzero. Choose a two-coloring c of $\widetilde{Q}^{(1)}$. Then $c \circ \tau = c + 1$.*

For $f \in F(Q)$, choose a lift $\widetilde{f}$ and write its positively oriented boundary, beginning at a color-zero vertex, as $\widetilde{A} - \widetilde{C} - \widetilde{B} - \widetilde{D} - \widetilde{A}$. Then $\chi_\beta(f) := \chi_\beta(A, B; C, D)$ is independent of the chosen lift and of the chosen color-zero initial vertex.

Proof. The identity $c \circ \tau = c + 1$ follows from Proposition 4.3. Starting at the other color-zero vertex replaces $\chi_\beta(A, B; C, D)$ by $\chi_\beta(B, A; D, C)$, which has the same value.

The deck involution reverses the tautological orientation and exchanges the colors. Hence the positively oriented boundary of $\tau\widetilde{f}$, beginning at a color-zero vertex, is $\tau\widetilde{C} - \tau\widetilde{A} - \tau\widetilde{D} - \tau\widetilde{B} - \tau\widetilde{C}$. It gives $\chi_\beta(C, D; A, B) = \chi_\beta(A, B; C, D)$ by the symmetry of β. □

Theorem 4.5 (Polarity master theorem)**.** *Let $Q^{\mathcal{L}}$ be an orientation-twisted polarity tiling of a closed connected surface S, and assume that every edge pairing is nonzero.*

If S is orientable, choose an orientation of S and a two-coloring of $Q^{(1)}$, and read $\chi_\beta(f)$ from the positively oriented boundary of f beginning at a color-zero vertex. If S is nonorientable, define $\chi_\beta(f)$ by Lemma 4.4. Then

$$\prod_{f \in F(Q)} \chi_\beta(f) = 1.$$

Consequently, if every face except a marked face f_0 is coherent, then f_0 is coherent.

Proof. In the orientable case, the equality $\ell = w_{1,S} = 0$ implies that $Q^{(1)}$ is bipartite. The result is then the usual Fomin–Pylyavskyy cancellation argument.

Indeed, in either case an edge pairing occurs with exponent $+1$ when the positively oriented face boundary traverses the edge from color zero to color one, and with exponent -1 in the opposite direction. The two faces incident with an edge induce opposite boundary directions. In the nonorientable case, compute their cross ratios using the two lifts incident with the same lifted edge; this is permitted by Lemma 4.4. Thus every edge pairing occurs once with each exponent and cancels. The product identity follows, and it forces $\chi_\beta(f_0) = 1$ when all other faces are coherent. □

Example 4.6 (The Hesse polarity tiling)**.** The Hesse configuration gives a polarity tiling of $\mathbb{RP}^2$. Under the edge-admissibility hypotheses of Theorem 4.5, coherence of all but one face forces coherence of the remaining face. The geometric configuration and its projective-plane tiling are shown in Figure 10.

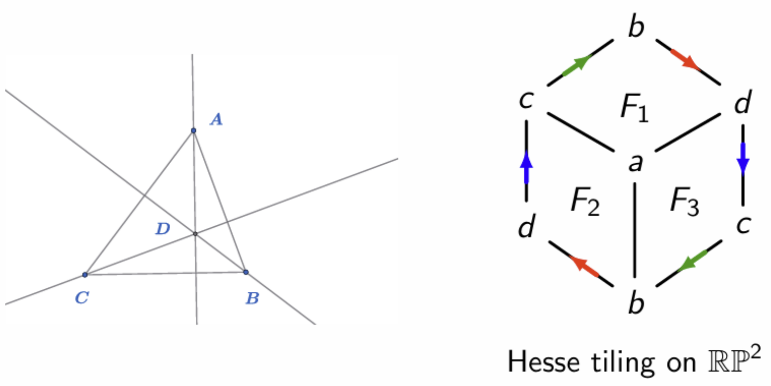


Figure 10: The Hesse configuration and its polarity tiling on $\mathbb{RP}^2$. Repeated boundary labels and colored arrows encode the edge identifications.

Example 4.7 (A Klein bottle polarity tiling)**.** Let $A_1, A_2, A_3, B_1, B_2, B_3$ be points in the polarized projective plane. For $\{i, j, k\} = \{1, 2, 3\}$, set

$$a_i = A_j A_k, \qquad b_i = B_j B_k, \qquad c_i = A_i B_i.$$

Consider the six conditions

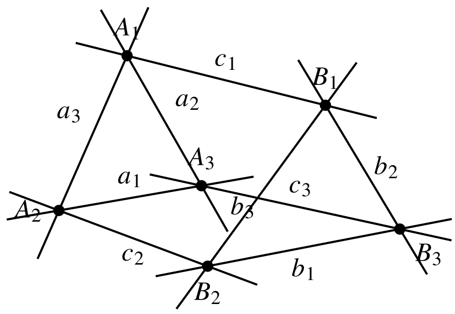
$$a_i \perp_\beta c_i, \qquad b_i \perp_\beta c_i \qquad (i = 1, 2, 3).$$

Under the nondegeneracy assumptions of Theorem 4.5, any five of these conditions imply the sixth.

Let $C_i = a_i \cap b_i$. Then $c_i \perp_\beta C_i$ for $i \in \{1, 2, 3\}$. If C_1, C_2, C_3 degenerate to a collinear triple, or c_1, c_2, c_3 degenerate to a concurrent triple, the same example specializes to the self-dual Desargues configuration. See Figure 11.

$$\begin{tikzcd}
B_2 \arrow[r] \arrow[d] & A_1 \arrow[r] \arrow[d, no head] & A_3 \arrow[r] \arrow[d, no head] & B_2 \arrow[d] \\
B_1 \arrow[r, no head] \arrow[d] & B_3 \arrow[r, no head] \arrow[d, no head] & A_2 \arrow[r, no head] \arrow[d, no head] & A_1 \arrow[d] \\
A_3 \arrow[r] & B_2 \arrow[r] & B_1 \arrow[r] & A_3
\end{tikzcd}$$

(a) The Klein bottle polarity tiling.



(b) The geometric realization.

Figure 11: A Klein bottle polarity tiling and its geometric realization. Boundary edges with matching endpoint labels and orientations are identified.

Remark 4.8 (The bordered case). The same cancellation applies to a compact connected surface with boundary. Each boundary component B_j has an orientable collar, so $\langle w_{1,S}, [B_j] \rangle = 0$. Since $\ell = w_{1,S}$, its length is even; write it as $2n_j$.

In the orientable case, use the chosen orientation and coloring. In the strictly orientation-twisted case, choose either lift of B_j, give it the boundary orientation induced from the tautological orientation of the cover, and begin at a color-zero vertex. The resulting cyclic labeling determines a polarity $2n_j$-gon

$$P_j = A_{j,1} C_{j,1} \cdots A_{j,n_j} C_{j,n_j}.$$

The choice of lift does not affect $\chi_\beta(P_j)$, and cancellation of the interior edge pairings gives

$$\prod_{f \in F(Q)} \chi_\beta(f) = \prod_{j=1}^{b} \chi_\beta(P_j).$$

In particular, if S has one boundary component and every face is coherent, then its boundary is a coherent polarity $2n$-gon.

Putting $\ell_{j,i} = C_{j,i}^\perp$ identifies this polarity $2n_j$-gon with the alternately labeled polygon $A_{j,1}\ell_{j,1} \cdots A_{j,n_j}\ell_{j,n_j}$ of Fomin and Pylyavskyy. Under the hypotheses of their Proposition 4.2, its coherence is equivalent to tileability by coherent quadrilateral tiles, yielding the corresponding one-boundary statement of their Corollary 4.3 [2, Proposition 4.2 and Corollary 4.3].

Remark 4.9. The same cancellation applies to a compact connected surface with boundary. Each boundary component B_j has an orientable collar, so $\langle w_{1,S}, [B_j] \rangle = 0$. Since $\ell = w_{1,S}$, its length is even; write it as $2n_j$.

In the orientable case, use the chosen orientation and coloring. In the strictly orientation-twisted case, choose either lift of B_j, give it the boundary orientation induced from the tautological orientation of the cover, and begin at a color-zero vertex. The resulting cyclic labeling determines a polarity $2n_j$-gon

$$P_j = A_{j,1}C_{j,1}\cdots A_{j,n_j}C_{j,n_j}.$$

The choice of lift does not affect $\chi_\beta(P_j)$, and cancellation of the interior edge pairings gives

$$\prod_{f\in F(Q)} \chi_\beta(f) = \prod_{j=1}^{b} \chi_\beta(P_j).$$

In particular, if S has one boundary component and every face is coherent, then its boundary is a coherent polarity $2n$-gon.

Putting $\ell_{j,i} = C_{j,i}^{\perp}$ identifies this polarity $2n_j$-gon with the alternately labeled polygon $A_{j,1}\ell_{j,1}\cdots A_{j,n_j}\ell_{j,n_j}$ of Fomin and Pylyavskyy. Under the hypotheses of their Proposition 4.2, its coherence is equivalent to tileability by coherent quadrilateral tiles, yielding the corresponding one-boundary statement of their Corollary 4.3 [2, Proposition 4.2 and Corollary 4.3].

5 Self-Duality, Lifting, and Descent

5.1 Lift to Self-dual Incidence Theorems

Definition 5.1 (Maps of tilings)**.** Let $Q^{\mathcal{B}}$ and $(Q')^{\mathcal{B}'}$ be bipartite tilings. A map of tilings

$$(f,\sigma) : Q^{\mathcal{B}} \longrightarrow (Q')^{\mathcal{B}'}$$

consists of a cellular map $f : Q \to Q'$ of quadrangular complexes and a bijection $\sigma : \mathbb{N} \to \mathbb{N}$ such that

$$\mathcal{B}' \circ f = (\mathrm{id}_{\mathbb{F}_2}, \sigma) \circ \mathcal{B}.$$

It is an isomorphism of tilings if f is a cellular isomorphism.

Let $\varphi(b,n) = (b+1,n)$, and write $\varphi\mathcal{B} := \varphi \circ \mathcal{B}$.

Proposition 5.2 (Color swapping and duality)**.** *Suppose that*

$$(\tau,\sigma) : Q^{\mathcal{B}} \xrightarrow{\sim} Q^{\varphi\mathcal{B}}$$

is an isomorphism of tilings whose underlying cellular map τ is an involution. Then projective duality identifies $I(f)^\vee$ with $I(\tau f)$ for every face f, up to the numerical relabeling σ.

Consequently, if $f \neq \tau f$, the pair-marked statement carried by $Q^{\mathcal{B}}_{f,\tau f}$ is self-dual.

Proof. If $\mathcal{B}(v) = (b,n)$, the defining equivariance of the tiling isomorphism gives

$$\mathcal{B}(\tau v) = (b+1, \sigma(n)).$$

Thus τ exchanges point and line types and relabels the numerical index by σ. Projective duality therefore carries the face relation $I(f)$ to $I(\tau f)$.

The involution τ permutes the faces outside $\{f, \tau f\}$ and exchanges the two marked faces. It consequently preserves the premises of the pair-marked statement and interchanges its two conclusions. $\square$

Theorem 5.3 (Canonical lift)**.** *Let $Q_f^{\mathcal{L}}$ be a strictly orientation-twisted polarity tiling, and let $(\widetilde{Q}, \tau)$ be its orientation-cover lift. Choose a two-coloring c of $\widetilde{Q}^{(1)}$ and define*

$$\widetilde{\mathcal{B}}(\widetilde{v}) := \big(c(\widetilde{v}), \mathcal{L}(\pi(\widetilde{v}))\big).$$

For either lift $\widetilde{f}$ of f, the canonical lift of $Q_f^{\mathcal{L}}$ is the pair-marked tiling

$$\widetilde{Q_f^{\mathcal{L}}} := \widetilde{Q}^{\widetilde{\mathcal{B}}}_{\widetilde{f}, \tau\widetilde{f}}.$$

The deck involution exchanges the marked faces and induces an involutive isomorphism of tilings

$$\tau : \widetilde{Q}^{\widetilde{\mathcal{B}}} \xrightarrow{\sim} \widetilde{Q}^{\varphi\widetilde{\mathcal{B}}}.$$

Hence the associated pair-marked statement is valid and self-dual.

Proof. By Proposition 4.3, $c \circ \tau = c + 1$. Since $\mathcal{L} \circ \pi$ is deck-invariant,

$$\widetilde{\mathcal{B}} \circ \tau = \varphi \circ \widetilde{\mathcal{B}}.$$

Equivalently, $(\varphi\widetilde{\mathcal{B}}) \circ \tau = \widetilde{\mathcal{B}}$, so τ is an isomorphism of tilings in the sense of Definition 5.1. It exchanges $\widetilde{f}$ and $\tau\widetilde{f}$.

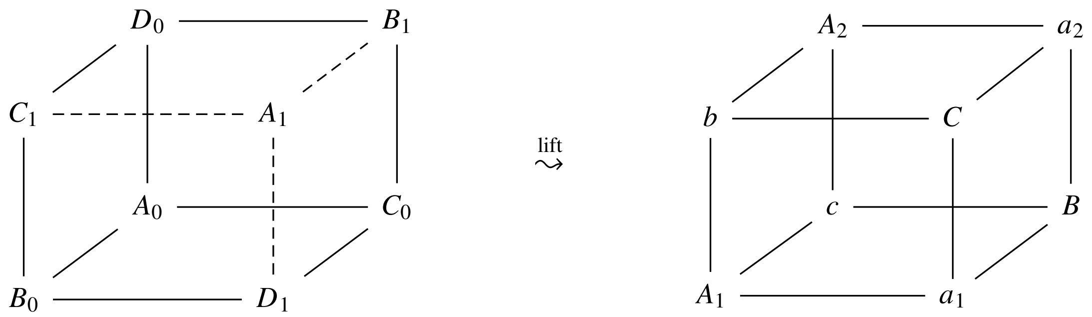
The ordinary master theorem gives the pair-marked implication carried by $\widetilde{Q}^{\widetilde{\mathcal{B}}}_{\widetilde{f}, \tau\widetilde{f}}$, while Proposition 5.2 shows that this implication is self-dual. $\square$

Example 5.4 (The Hesse–Desargues lift)**.** The hemicube quadrangulation of $\mathbb{RP}^2$ lifts to the cube quadrangulation of S^2. At the level of incidence statements, this realizes the classical passage from Hesse's theorem to Desargues' theorem [1].

If f is the conclusion face of the hemicube, then $\pi^{-1}(f) = \{\widetilde{f}, \tau\widetilde{f}\}$ consists of two antipodal faces, and the canonical lift is the pair-marked cube

$$\widetilde{Q}_{\widetilde{f}, \tau\widetilde{f}}.$$

The one-boundary Hesse tiling therefore lifts to a two-boundary Desargues tiling whose conclusions are exchanged by the deck involution and, accordingly, by projective duality; see Figure 12.



(a) The Hesse hemicube.

(b) The Desargues cube.

Figure 12: The orientation-cover lift from the Hesse hemicube to the Desargues cube.

5.2 Descent by an Admissible Involution

We now consider the converse direction of Theorem 5.3, which we call descent.

Definition 5.5 (Admissible quadrangulation involution)**.** Let $\widetilde{Q}$ be a connected bipartite quadrangulation of a closed orientable surface $\widetilde{S}$, with two-coloring c. A cellular involution τ is *admissible* if

(i) it is fixed-point-free on $|\widetilde{Q}| = \widetilde{S}$;

(ii) it reverses the orientation of $\widetilde{S}$;

(iii) it exchanges the colors: $c \circ \tau = c + 1$;

(iv) the orbit cell structure $\widetilde{Q}/\langle\tau\rangle$ is a regular quadrangulation.

The final condition is automatic for a lifted quadrangulation, but is stated explicitly in the converse: topological freeness alone does not rule out identifications between distinct boundary portions of a chosen cell representative.

Theorem 5.6 (Cover characterization and labeled descent)**.** *A connected bipartite regular quadrangulation $\widetilde{Q}$ is the orientation double cover of a closed nonorientable orientation-twisted quadrangulation if and only if it admits an admissible involution.*

Suppose in addition that $\widetilde{Q}$ carries a polarity-compatible labeling $\widetilde{\mathcal{B}} = (c, n)$ satisfying$\widetilde{\mathcal{B}} \circ \tau = \varphi \circ \widetilde{\mathcal{B}}$. Then $\mathcal{L}([\widetilde{v}]) := n(\widetilde{v})$ is a well-defined polarity labeling on the quotient, and $\widetilde{Q}^{\widetilde{\mathcal{B}}}/\langle\tau\rangle = Q^{\mathcal{L}}$.

Proof. If $\widetilde{Q}$ is the orientation cover of such a Q, the deck involution is fixed-point-free and orientation-reversing. The color-exchange property follows from Proposition 4.3, and the quotient cell structure is the original regular quadrangulation.

Conversely, let τ be admissible. A fixed-point-free involution of a closed surface has a surface quotient, and the projection

$$\widetilde{S} \longrightarrow S := \widetilde{S}/\langle\tau\rangle$$

is a two-sheeted covering. The regularity clause makes the cellular quotient $Q := \widetilde{Q}/\langle\tau\rangle$ a regular quadrangulation. Since τ reverses orientation, S is nonorientable and this covering is its orientation cover.

It remains to prove that Q is orientation-twisted. Let γ be a closed edge path in Q and lift it from $\widetilde{v}$. If γ preserves orientation, its lift is closed and hence has even length in the bipartite graph

$\widetilde{Q}$. If γ reverses orientation, its lift ends at $\tau\widetilde{v}$, which has the opposite color, so its length is odd. By Proposition 4.3, $[\lambda] = w_{1,S}$.

Finally, the equivariance equation says pointwise that

$$(c(\tau\widetilde{v}), n(\tau\widetilde{v})) = (c(\widetilde{v}) + 1, n(\widetilde{v})).$$

Hence n is constant on vertex orbits and defines $\mathcal{L}$ downstairs. The polarity-compatible interpretation of type-one vertices shows that every orbit of a lifted ordinary tile descends to the corresponding polarity tile, which proves the labeled quotient identity. $\square$

If $\widetilde{S}$ has orientable genus g, then its quotient has nonorientable genus $g + 1$, since $\chi(\widetilde{S}) = 2\chi(S)$. The numbers of vertices, edges, and faces downstairs are half the corresponding numbers upstairs. For a fixed $\widetilde{Q}$, quotient quadrangulations together with an identification of their orientation cover with $\widetilde{Q}$ are classified by conjugacy classes of admissible involutions in $\mathrm{Aut}_{\mathrm{map}}(\widetilde{Q})$: changing the identification conjugates the deck involution, and an isomorphism of quotients lifts to such a conjugacy.

6 Division Rings and Gauge Generalization

6.1 Gauge connections on surface graphs

We first recall the gauge-theoretic mechanism used by Izosimov. Let Γ be a connected graph, let $\vec{E}(\Gamma)$ be its set of oriented edges, and write $\bar{e}$ for the reversal of e. For a group G, a G-connection is a map $U : \vec{E}(\Gamma) \to G$ satisfying $U(\bar{e}) = U(e)^{-1}$. Its holonomy along an edge path $\gamma = e_1 \cdots e_r$ is $\mathrm{Hol}_U(\gamma) = U(e_1) \cdots U(e_r)$. A vertex function $q : V(\Gamma) \to G$ acts by

$$(q \cdot U)(e) = q(t(e))U(e)q(h(e))^{-1}.$$

Connections in the same orbit are gauge-equivalent.

Suppose now that Γ is a map on a compact connected surface S, so the closures of the complementary components are the faces of a cellular decomposition. A connection is *flat* if its holonomy around every face is trivial. Changing the base vertex of a facial boundary conjugates its holonomy, while reversing the boundary orientation inverts it, so flatness is independent of these choices. This is Izosimov's terminology for connections on maps [4, Definitions 3.1–3.7 and Remark 3.8].

The following formulation combines results proved by Izosimov in Propositions 3.5, 3.9, and 3.10 [4].

Theorem 6.1 (Gauge correspondence, after Izosimov)**.** *Fix $v_0 \in V(\Gamma)$. Holonomy induces natural bijections*

$$\frac{\{G\text{-connections on } \Gamma\}}{\{\text{gauge equivalence}\}} \cong \mathrm{Hom}\big(\pi_1(\Gamma, v_0), G\big)/G$$

and

$$\frac{\{\text{flat } G\text{-connections on } \Gamma\}}{\{\text{gauge equivalence}\}} \cong \mathrm{Hom}\big(\pi_1(S, v_0), G\big)/G,$$

where G acts on representations by conjugation.

Proof. Immediate backtracking contributes $U(e)U(\bar{e}) = 1$, so based holonomy defines a representation of $\pi_1(\Gamma, v_0)$. A gauge transformation conjugates this representation by $q(v_0)$.

Conversely, choose a maximal tree $T \subset \Gamma$, and let p_v be the tree path from v_0 to v. Given $\rho : \pi_1(\Gamma, v_0) \to G$, define, for every oriented edge e,

$$U_\rho(e) := \rho\big([p_{t(e)} e\, p_{h(e)}^{-1}]\big).$$

This is a connection, it is trivial on T, and its holonomy is ρ. Every connection can be brought to this tree-normalized form by gauge, and the remaining constant gauge transformations give precisely conjugation by G. This proves the first bijection.

By van Kampen's theorem, the kernel of $\pi_1(\Gamma, v_0) \to \pi_1(S, v_0)$ is normally generated by the facial boundary loops. Hence a connection is flat exactly when its holonomy representation factors through $\pi_1(S, v_0)$, which gives the second bijection. □

Excision is the group-theoretic condition that turns flatness on the premise faces into flatness on the marked face.

Definition 6.2 (Group-valued excisability)**.** Let Q_{f_0} be a marked regular cellular quadrangulation and let G be a group. The marked face f_0 is *G-excisable* if every G-connection on $Q^{(1)}$ that is flat on every face $f \neq f_0$ is also flat on f_0.

Theorem 6.3 (Group-valued excision criterion)**.** *Let Q_{f_0} be an orientation-twisted marked regular quadrangulation of a closed connected surface S, and let G be a group. Put $Y := S \setminus \mathrm{Int}(f_0)$. Choose a boundary vertex $x_0 \in V(Q) \cap \partial f_0$ and an orientation of the boundary edge loop of f_0, and let $\delta_{f_0} \in \pi_1(Y, x_0)$ be its class. The following conditions are equivalent:*

(i) *The face f_0 is G-excisable.*

(ii) *Every homomorphism $\rho : \pi_1(Y, x_0) \to G$ satisfies $\rho(\delta_{f_0}) = 1$.*

Changing the direction of the boundary loop inverts δ_{f_0}, while changing the base vertex conjugates it. Consequently, condition (ii) *is independent of these choices; the base vertex will henceforth be suppressed.*

Proof. Assume first that f_0 is G-excisable, and let $\rho : \pi_1(Y, x_0) \to G$ be a homomorphism. The graph $Q^{(1)} \subset Y$ is a map whose faces are precisely the faces of Q distinct from f_0. By Izosimov's gauge correspondence, Theorem 6.1, there is a flat G-connection U on this map whose holonomy representation is conjugate to ρ. Regarded as a connection on $Q^{(1)}$, it is flat on every face other than f_0, and hence also on f_0 by excisability. Thus $\rho(\delta_{f_0}) = 1$, since conjugation does not affect whether an element is trivial.

Conversely, let U be a G-connection on $Q^{(1)}$ that is flat on every face distinct from f_0. Viewed as a flat connection on the map $Q^{(1)} \subset Y$, its graph holonomy factors through a homomorphism $\rho_U : \pi_1(Y, x_0) \to G$ by [4, Proposition 3.9]. The holonomy of U around f_0 is $\rho_U(\delta_{f_0})$, up to inversion and conjugation, and is therefore trivial by (ii). Hence f_0 is G-excisable. □

For an ordinary bipartite tiling over a division ring, Izosimov orients each edge from its point vertex A_e to its hyperplane vertex ℓ_e and assigns the value $\ell_e(A_e) \in D^\times$. Projective rescaling acts by gauge, and Proposition 5.14 identifies tile coherence with trivial facial holonomy. Izosimov then combines this connection with his spherical flatness result, Corollary 3.11, to prove the spherical division-ring theorem [4, Corollary 3.11, Proposition 5.14, and the proof of Theorem 5.13]. The polarity setting retains this mechanism, but Hermitian symmetry requires a larger coefficient group.

6.2 The σ-Hermitian construction

Let D be a division ring, let V be a three-dimensional right D-vector space, and let $\sigma : D \to D$ be an involutive anti-automorphism:

$$\sigma(ab) = \sigma(b)\sigma(a), \qquad \sigma^2 = \mathrm{id}.$$

A nondegenerate σ-Hermitian form is a map $h : V \times V \to D$ satisfying

$$h(xa, yb) = \sigma(a)h(x, y)b, \qquad h(y, x) = \sigma(h(x, y)).$$

It defines the projective polarity $[x] \mapsto [x]^{\perp_h}$, where $[x]^{\perp_h} = \{[y] : h(x, y) = 0\}$. These conventions are standard for Hermitian forms over division rings [6, Chapter 8]. The right-module convention is important: semilinearity in the first variable forces σ to reverse multiplication.

Define

$$\widehat{\sigma} : D^\times \longrightarrow D^\times, \qquad \widehat{\sigma}(g) = \sigma(g)^{-1}.$$

Since

$$\widehat{\sigma}(gk) = \big(\sigma(k)\sigma(g)\big)^{-1} = \widehat{\sigma}(g)\widehat{\sigma}(k), \qquad \widehat{\sigma}^2 = \mathrm{id},$$

this is an involutive group automorphism. Following the manuscript notation, put

$$D^\times_\sigma := D^\times \rtimes_{\widehat{\sigma}} \mathbb{F}_2, \qquad (g, a)(k, b) = \big(g\widehat{\sigma}^a(k), a + b\big). \tag{1}$$

Let $p : D^\times_\sigma \to \mathbb{F}_2$ be the projection onto the second coordinate.

Fix a realization $\mathcal{R} : \mathbb{N} \to V \setminus \{0\}$ of the numerical labels. For every vertex v, put $x_v = (\mathcal{R} \circ \mathcal{L})(v)$ and $H(v, w) := h(x_v, x_w)$. Assume $H(v, w) \neq 0$ on every edge.

Proposition 6.4 (Hermitian connection and face holonomy)**.** *On an oriented edge (v, w), define*

$$U_h(v, w) := \big(H(v, w), \lambda(vw)\big) = \big(H(v, w), 1\big) \in D^\times_\sigma.$$

Then $U_h(w, v) = U_h(v, w)^{-1}$, so U_h is a $D^\times_\sigma$-connection. For every edge path γ,

$$p\big(\mathrm{Hol}_{U_h}(\gamma)\big) = \lambda(\gamma) = |\gamma| \pmod 2.$$

In particular, if Q is orientation-twisted, then every closed edge path γ satisfies

$$p\big(\mathrm{Hol}_{U_h}(\gamma)\big) = \langle w_{1,S}, [\gamma]\rangle.$$

For a face with boundary $A - C - B - D - A$,

$$\mathrm{Hol}_{U_h}(\partial f) = \big(\chi_h(A, B; C, D), 0\big), \tag{2}$$

where the ordered noncommutative polarity cross ratio is

$$\chi_h(A, B; C, D) := H(A, C)H(B, C)^{-1}H(B, D)H(A, D)^{-1}. \tag{3}$$

Changing projective representatives changes U_h by gauge and conjugates χ_h. Consequently, the condition $\chi_h = 1$, equivalently flatness on the face, is projectively well-defined.

Proof. In the semidirect product,
$$(g,1)^{-1} = (\sigma(g),1).$$

Hermitian symmetry therefore gives

$$U_h(w,v) = \big(\sigma(H(v,w)),1\big) = U_h(v,w)^{-1}.$$

If a representative x_v is replaced by $x_v a_v$, then $H'(v,w) = \sigma(a_v)H(v,w)a_w$. The vertex function $q(v) = (\sigma(a_v),0)$ satisfies

$$(q \cdot U_h)(v,w) = q(v)U_h(v,w)q(w)^{-1} = U'_h(v,w),$$

so projective rescaling is gauge.

The second coordinate is additive under multiplication in $D^{\times}_{\sigma}$, and every edge has $\lambda(e) = 1$. This proves the path-parity formula. For an orientation-twisted quadrangulation, $[\lambda] = w_{1,S}$, giving the assertion for closed paths.

Multiplying the four edge values in boundary order gives

$$\begin{aligned}&(H(A,C),1)(H(C,B),1)(H(B,D),1)(H(D,A),1)\\ &\quad = \big(H(A,C)\widehat{\sigma}(H(C,B))H(B,D)\widehat{\sigma}(H(D,A)),0\big).\end{aligned}$$

Hermitian symmetry gives

$$H(C,B) = \sigma(H(B,C)), \qquad H(D,A) = \sigma(H(A,D)),$$

so the first coordinate is (2). For rescalings $x_A a, x_B b, x_C c, x_D d$, let χ'_h denote the resulting expression. Cancellation in the displayed order yields

$$\chi'_h(A,B;C,D) = \sigma(a)\chi_h(A,B;C,D)\sigma(a)^{-1}.$$

This proves all assertions. □

Proposition 6.5 (Geometric coherence over a division ring)**.** *Assume that A,B are projectively independent, that C,D are projectively independent, and that the four edge pairings in* (3) *are nonzero. Then*
$$\chi_h(A,B;C,D) = 1 \iff (AB)^{\perp_h} \in CD.$$

Here $(AB)^{\perp_h}$ is the one-dimensional right subspace orthogonal to the right span of A,B.

Proof. Write $A = [x_A]$, $B = [x_B]$, $C = [x_C]$, and $D = [x_D]$. The point $(AB)^{\perp_h}$ lies on CD exactly when there are right scalars x,y, not both zero, such that $x_C x + x_D y$ is orthogonal to both x_A and x_B, or equivalently
$$H(A,C)x + H(A,D)y = 0, \qquad H(B,C)x + H(B,D)y = 0.$$

Nonvanishing forces $y \neq 0$, and the first equation gives $x = -H(A,C)^{-1}H(A,D)y$. Substitution into the second gives
$$H(B,D) = H(B,C)H(A,C)^{-1}H(A,D),$$

which, after multiplying in the permitted order, is equivalent to $\chi_h = 1$. □

We write $AB \perp_h CD$ for the equivalent conditions in Proposition 6.5. Accordingly, a face with boundary $A - C - B - D - A$ is called h-coherent if $AB \perp_h CD$.

Thus $h \mapsto U_h$ is a direct extension of the symmetric bilinear construction: the local incidence condition is still exactly flatness, but noncommutativity forces both the order in (3) and the parity component in $D_\sigma^\times$.

6.3 The Hermitian polarity master theorem

Theorem 6.6 (Hermitian polarity master theorem over division rings)**.** *Let $Q_{f_0}^{\mathcal{L}}$ be an orientation-twisted marked polarity tiling of a closed connected surface S, and let $(h, \mathcal{R})$ be a σ-Hermitian realization with nonzero edge pairings. Assume that the two opposite vertex pairs of every face are projectively independent. Suppose that every face other than f_0 is h-coherent. Then f_0 is h-coherent in each of the following cases:*

(i) *$S \cong S^2$, with D and σ arbitrary;*

(ii) *$S \cong \Sigma_g$ for $g \geq 1$, with D commutative and σ arbitrary;*

(iii) *$S \cong N_k$ for $k \geq 1$, with D commutative and $\sigma = \mathrm{id}$.*

Proof. Put $Y = S \setminus \mathrm{Int}(f_0)$, and let $i : Y \hookrightarrow S$ be the inclusion. By Propositions 6.4 and 6.5, the connection U_h is flat on every face of Y. It therefore determines a representation $\rho : \pi_1(Y) \to D_\sigma^\times$. The path-parity formula and the orientation-twisted identity $[\lambda] = w_{1,S}$ show that $p(\rho(\gamma)) = \langle i^* w_{1,S}, [\gamma] \rangle$ for every $\gamma \in \pi_1(Y)$. The holonomy around f_0 is $\rho(\delta_{f_0})$, up to inversion and conjugation, so it remains to prove that this element is trivial. We use the standard boundary words for punctured surfaces [3, Section 1.2].

If $S = S^2$, then Y is a disk and $\delta_{f_0} = 1$. This proves (i).

Let $S = \Sigma_g$ with $g \geq 1$. Since S is orientable, $w_{1,S} = 0$, so the image of ρ is contained in the even subgroup $D^\times \times \{0\}$. Write $\rho(a_i) = (\alpha_i, 0)$ and $\rho(b_i) = (\beta_i, 0)$. Since $\delta_{f_0} = \prod_{i=1}^{g} [a_i, b_i]$, the orientable boundary word gives

$$\rho(\delta_{f_0}) = \left(\prod_{i=1}^{g} [\alpha_i, \beta_i], 0 \right).$$

When D is commutative, every commutator in $D^\times$ is trivial, which proves (ii). Notice that no condition on σ is needed: only the even subgroup occurs.

Finally, let $S = N_k$. The standard generators x_j are orientation-reversing, hence $\rho(x_j) = (q_j, 1)$ for some $q_j \in D^\times$, and $\delta_{f_0} = \prod_{j=1}^{k} x_j^2$. From (1),

$$(q_j, 1)^2 = \big(q_j \widehat{\sigma}(q_j), 0\big) = \big(q_j \sigma(q_j)^{-1}, 0\big).$$

If $\sigma = \mathrm{id}$, each of these squares is the identity. The nonorientable boundary word therefore gives $\rho(\delta_{f_0}) = 1$, proving (iii). Since the identity map can be an anti-automorphism only on a commutative division ring, the commutativity assumption in (iii) is also automatic from $\sigma = \mathrm{id}$.

In all three cases U_h is flat around f_0. Its face holonomy is therefore trivial, and Propositions 6.4 and 6.5 imply that f_0 is h-coherent. □

Remark 6.7 (Restriction of converse). The gauge correspondence realizes every abstract representation by a connection on the graph, but not necessarily by a connection of the form U_h. Such a connection is constrained by a single Hermitian Gram matrix, by the repeated numerical labels of the tiling, and by the rank of the ambient projective space. In Izosimov's ordinary point–hyperplane setting, the converse construction assigns point vectors and hyperplane covectors independently and requires additional simplicity and dimension hypotheses [4, Theorem 5.17 and its second proof]. Those choices are unavailable for a polarity realization, where a hyperplane covector must be induced from the same form h. Consequently, an abstract representation with nontrivial boundary holonomy does not by itself provide a Hermitian polarity counterexample; a separate Hermitian realization theorem would be needed. Theorem 6.6 therefore makes no converse claim.

Example 6.8 (Failure of the complex Hermitian Hesse analogue)**.** Let

$$D = \mathbb{C}, \qquad \sigma(z) = \overline{z}, \qquad h(x, y) = \overline{x}^T y,$$

and take

$$A = (1, 0, 0), \quad B = (1, 1, 0), \quad C = (1, i, 1), \quad D = (1, i, -1 - i).$$

All six pairings between distinct points are nonzero. For the three opposite-line conditions in the Hesse configuration of Example 5.4, the relevant cross ratios are

$$\begin{aligned}
\chi_h(A, B; C, D) &= 1\,(1 + i)^{-1}(1 + i)\,1^{-1} = 1,\\
\chi_h(A, C; B, D) &= 1\,(1 - i)^{-1}(1 - i)\,1^{-1} = 1,\\
\chi_h(A, D; B, C) &= 1\,(1 - i)^{-1}(1 + i)\,1^{-1} = i \neq 1.
\end{aligned}$$

Thus $AB \perp_h CD$ and $AC \perp_h BD$, but $AD \not\perp_h BC$. This is an explicit h-realizable obstruction, not merely an abstract representation-theoretic one. Equivalently, deleting the third hemicube face gives a Möbius band with $\delta_{f_0} = x^2$. For the odd core generator $x = [A\,B\,C\,A]$, direct multiplication gives

$$\rho(x) = (1, 1)(1 + i, 1)(1, 1) = \big((1 + i)/2, 1\big).$$

Writing $p = (1 + i)/2$, we obtain

$$\rho(\delta_{f_0}) = (p\overline{p}^{-1}, 0) = (i, 0) \neq 1.$$

7 Tiling Surface

Fix a formal polarity incidence presentation $I : Q_1, \ldots, Q_m \Longrightarrow Q_0$. We retain the notation X_I, G_I, and δ_I, and the definition of a surface-tiling diagram, from the general setting. Throughout this section the conclusion satisfies the standing nondegeneracy hypothesis, and only elementary fixed-form tiling proofs are considered. The argument follows Zhang's van Kampen construction for elementary tilings [7, Lemmas 3.3–3.4 and Theorem 3.6], with an additional parity constraint that detects nonorientability.

7.1 The orientable model and parity-refined lengths

In the ordinary bipartite setting, Zhang proves that the least orientable genus of an elementary tiling proof is

$$g_{\mathrm{el}}(I) = \mathrm{cl}_{G_I}(\delta_I). \tag{4}$$

The proof first reads the standard boundary word of a punctured orientable surface as a product of commutators and then reverses the construction by gluing paired arcs in a van Kampen disk [7, Theorem 1.9 and Section 3].

Let λ be the all-ones cochain on $X_I^{(1)}$. Every premise attaching word has four edge occurrences, so $d\lambda = 0$. We use the same symbol $\ell \in H^1(X_I; \mathbb{F}_2)$ for the resulting cohomology class and $\ell : G_I \to \mathbb{F}_2$ for the induced parity character on the conclusion component. It evaluates on a closed edge path as its length modulo two. The conclusion word also has four edge occurrences, hence $\ell(\delta_I) = 0$.

Definition 7.1 (Parity-refined factorization lengths)**.** For $\gamma \in \ker \ell$, set

$$\mathrm{sl}^{\mathrm{odd}}_{G_I}(\gamma) := \min\left\{k \;\middle|\; \gamma = z_1^2 \cdots z_k^2,\ \ell(z_j) = 1 \text{ for all } j\right\}, \tag{5}$$

$$\mathrm{cl}^{\mathrm{even}}_{G_I}(\gamma) := \min\left\{g \;\middle|\; \gamma = [a_1, b_1] \cdots [a_g, b_g],\ \ell(a_i) = \ell(b_i) = 0 \text{ for all } i\right\}. \tag{6}$$

Empty products are allowed, and a minimum is ∞ when no such factorization exists.

These lengths depend only on the conjugacy-and-inversion class of γ. Indeed, conjugation conjugates every factor and preserves its parity. Inversion reverses the order of the factors and uses $(z^2)^{-1} = (z^{-1})^2$ and $[a, b]^{-1} = [b, a]$; parity is again unchanged. In particular, both lengths are well-defined on δ_I.

Lemma 7.2 (Nondegenerate conclusion boundary)**.** *Let c_0 be the edge loop read on ∂Q_0. Under the standing formal nondegeneracy hypothesis, its four sides represent a nonzero class $[c_0]$ in $H_1(X_I^{(1)}; \mathbb{F}_2)$. Consequently, a van Kampen filling of either*

$$\left(\prod_{i=1}^{g} [\alpha_i, \beta_i]\right) c_0^{-1} \qquad \text{or} \qquad \zeta_1\zeta_1 \cdots \zeta_k\zeta_k c_0^{-1}$$

must contain at least one premise two-cell.

Proof. The commutator part abelianizes to zero, while $[\zeta_j\zeta_j] = 2[\zeta_j] = 0$ in $H_1(X_I^{(1)}; \mathbb{F}_2)$. Thus either displayed loop has the nonzero homology class $[c_0]$, up to the immaterial sign, and cannot be filled inside the one-skeleton. Any van Kampen filling over X_I must therefore use a premise two-cell. This is the mod-two analogue of Zhang's nondegeneracy argument [7, Lemma 3.3]. □

7.2 Orientable and nonorientable branches

For a surface-tiling diagram with underlying cellular map F, the orientation-twisted condition is $F^*\ell = w_1$. Write $\Sigma_{g,1}$ and $N_{k,1}$ for, respectively, the orientable genus-g and nonorientable genus-k surfaces with one boundary component.

Theorem 7.3 (Parity-refined tiling criterion)**.** *The following statements hold.*

(i) *For $g \geq 0$, an orientation-twisted orientable surface-tiling diagram for I of genus at most g exists if and only if $\mathrm{cl}^{\mathrm{even}}_{G_I}(\delta_I) \leq g$.*

(ii) *For $k \geq 1$, an orientation-twisted nonorientable surface-tiling diagram for I on $N_{k,1}$ exists if and only if δ_I admits a factorization*

$$\delta_I = z_1^2 \cdots z_k^2, \qquad \ell(z_j) = 1 \quad (1 \leq j \leq k). \tag{7}$$

Proof. Suppose first that a diagram has source $\Sigma_{r,1}$, where $r \le g$. For standard generators a_i, b_i, the boundary relation is $[\partial\Sigma_{r,1}] = \prod_{i=1}^{r}[a_i, b_i]$, up to conjugacy and inversion. Since $w_1(\Sigma_{r,1}) = 0$, the equality $F^*\ell = w_1$ gives $\ell(F_*a_i) = \ell(F_*b_i) = 0$. Applying F_* to the boundary relation yields an even-commutator factorization of δ_I, and therefore $\mathrm{cl}_{G_I}^{\mathrm{even}}(\delta_I) \le r \le g$.

Conversely, put $r = \mathrm{cl}_{G_I}^{\mathrm{even}}(\delta_I) \le g$ and choose a minimal factorization $\delta_I = \prod_{i=1}^{r}[A_i, B_i]$ with every A_i, B_i ℓ-even. If $r > 0$, no commutator in this factorization is trivial; choose reduced based edge-loop representatives α_i, β_i. For $r = 0$, there are no paired arcs. Choose also a based representative c_0 of the conclusion boundary. The loop

$$W_{\mathrm{or}} = \left(\prod_{i=1}^{r}[\alpha_i, \beta_i]\right) c_0^{-1}$$

is null-homotopic in X_I, so choose a van Kampen disk over the premise cells with boundary word W_{or}. Parametrize the displayed arcs by their boundary directions. Identify the point with parameter t on each α_i-arc with the point with parameter $1 - t$ on the corresponding $\bar{\alpha}_i$-arc, and do the same for β_i. Paired points have the same image in X_I, so the disk map descends. These boundary-orientation-reversing pairings are the standard handle identifications, and the quotient is a surface-tiling diagram on $\Sigma_{r,1}$, with remaining boundary c_0^{-1}; reversal of the conclusion boundary is allowed. The classes represented by the paired arcs form the standard basis of $H_1(\Sigma_{r,1}; \mathbb{F}_2)$, and their images are ℓ-even. Hence $F^*\ell = 0 = w_1(\Sigma_{r,1})$. This is Zhang's handle-gluing construction [7, Lemma 3.4 and Theorem 3.6].

Now let the source be $N_{k,1}$, with standard one-sided generators $x_1, \ldots, x_k$. They form a basis of $H_1(N_{k,1}; \mathbb{F}_2)$, satisfy $w_1(x_j) = 1$, and give the boundary relation $[\partial N_{k,1}] = x_1^2 \cdots x_k^2$, again up to conjugacy and inversion. An orientation-twisted diagram therefore gives $\delta_I = \prod_{j=1}^{k} F_*(x_j)^2$ with $\ell(F_*x_j) = 1$, proving the forward implication in (ii).

Conversely, assume (7). Choose based edge loops ζ_j representing z_j, and take a van Kampen disk for

$$W_{\mathrm{non}} = \zeta_1\zeta_1 \cdots \zeta_k\zeta_k c_0^{-1}.$$

For each j, identify points with the same parameter t on the two equally traversed ζ_j-arcs. Again the disk map descends pointwise. These boundary-orientation-preserving pairings are the standard crosscap identifications. Since $\ell(z_j) = 1$, every ζ_j is nonconstant, and the quotient is $N_{k,1}$, with remaining boundary c_0^{-1}. On the standard mod-two homology basis, both $F^*\ell$ and w_1 take the value one; hence they are equal. This is the nonorientable analogue of Zhang's van Kampen gluing.

Lemma 7.2 ensures in both constructions that the disk contains premise cells. Each of its two-cells maps homeomorphically to a premise square, as required by the definition of a surface-tiling diagram. Capping the remaining boundary with the conclusion square gives the corresponding closed tiling proof. Let $\widehat{S}$ be the capped surface and let $i : S \hookrightarrow \widehat{S}$ be the inclusion. The all-ones class on S extends across Q_0 because it evaluates trivially on its four-edge boundary. Naturality gives $i^*w_1(\widehat{S}) = w_1(S)$, and $i^* : H^1(\widehat{S}; \mathbb{F}_2) \to H^1(S; \mathbb{F}_2)$ is an isomorphism. Hence the extended parity class equals $w_1(\widehat{S})$, so the capped tiling is orientation-twisted. □

Remark 7.4. The word "exactly" in part (ii) is essential. Any positive-length odd-square factorization can be enlarged by two terms, by appending $u^2(u^{-1})^2$ for an odd factor u, but it need not be enlarged by one. Thus $\mathrm{sl}_{G_I}^{\mathrm{odd}}(\delta_I) \le k$ alone does not characterize the existence of a diagram on the fixed surface $N_{k,1}$.

7.3 Elementary Euler genus

Only after separating the two surface types do we combine them. For $\gamma \in \ker \ell$, define

$$\mathrm{tl}_{G_I}(\gamma) := \min \left\{ \mathrm{sl}^{\mathrm{odd}}_{G_I}(\gamma), 2\,\mathrm{cl}^{\mathrm{even}}_{G_I}(\gamma) \right\}. \tag{8}$$

Let $\mathrm{eg}_{\mathrm{el}}(I)$ be the least Euler genus of a closed surface supporting an elementary orientation-twisted tiling proof of I, with value ∞ if no such proof exists.

Theorem 7.5 (Elementary Euler-genus formula)**.** *For every fixed presentation I,*

$$\mathrm{eg}_{\mathrm{el}}(I) = \mathrm{tl}_{G_I}(\delta_I) = \min \left\{ \mathrm{sl}^{\mathrm{odd}}_{G_I}(\delta_I), 2\,\mathrm{cl}^{\mathrm{even}}_{G_I}(\delta_I) \right\}. \tag{9}$$

Moreover, any finite shortest factorization, together with a van Kampen disk for the corresponding boundary word, produces an optimal elementary tiling by the identifications in the proof of Theorem 7.3.

Proof. Delete the conclusion square from an elementary tiling proof. If its closed surface is orientable of genus g, Theorem 7.3(i) gives $\mathrm{cl}^{\mathrm{even}}_{G_I}(\delta_I) \le g$, while the Euler genus is $2g$. If the surface is nonorientable of genus k, part (ii) gives an exact k-term odd-square factorization, so $\mathrm{sl}^{\mathrm{odd}}_{G_I}(\delta_I) \le k$, and its Euler genus is k. This proves that every proof has Euler genus at least the right-hand side of (9).

If both factorization lengths are infinite, the forward implications of Theorem 7.3 exclude every elementary proof, so both sides of (9) are infinite. Otherwise, a finite shortest even-commutator factorization of length g produces a diagram on $\Sigma_{g,1}$, and hence a closed proof of Euler genus $2g$; a finite shortest odd-square factorization of positive length k produces a diagram on $N_{k,1}$, and hence a closed proof of Euler genus k. If $\delta_I = 1$ and the minimum is zero, the empty commutator factorization gives the sphere construction. Choosing a finite branch that realizes the minimum proves the reverse inequality and the construction statement. If the resulting surface had smaller Euler genus, the forward direction of the criterion would give a smaller value of the minimum in (9), a contradiction. □

Remark 7.6 (Recovery of Zhang's formula). In the ordinary bipartite setting every closed edge path is even, so $\ell = 0$, $\mathrm{cl}^{\mathrm{even}} = \mathrm{cl}$, and the positive-genus odd-square branch is absent. Formula (9) then becomes $\mathrm{eg}_{\mathrm{el}}(I) = 2\,\mathrm{cl}_{G_I}(\delta_I) = 2g_{\mathrm{el}}(I)$, which is precisely the Euler-genus form of (4).

Example 7.7 (The Hesse presentation)**.** For the hemicube presentation, X_I is the complement of the open conclusion face in the hemicube and hence a Möbius band. Thus

$$G_I \cong \mathbb{Z} = \langle x \rangle, \qquad \ell(x) = 1, \qquad \delta_I = x^2.$$

It follows that $\mathrm{sl}^{\mathrm{odd}}_{G_I}(\delta_I) = 1$. Since $\ker \ell = \langle x^2 \rangle$ is abelian and $x^2 \neq 1$, one has $\mathrm{cl}^{\mathrm{even}}_{G_I}(\delta_I) = \infty$. Therefore $\mathrm{eg}_{\mathrm{el}}(I) = 1$: an optimal surface-tiling diagram is the Möbius band, and capping its conclusion boundary gives $\mathbb{RP}^2$. No orientable elementary tiling proof exists. Its canonical orientation lift is an annulus whose two conclusion boundaries are exchanged by τ, as in Theorem 5.3.

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